

Quantum Computing Projects

List of Exam's Projects

- P1: Quantum Desk Calculator
- P2: Quantum Phase Estimation
- P3: Quantum Counting
- P4: Breaking Diffie-Hellman
- P5: Your Proposal

Instructions

The following list contains projects suggested for the exam of Quantum Computing. The student can choose one project to develop autonomously and complete by the date established for the exam. The mark of the project will be averaged with the one obtained in the written part to form the final mark.

Projects Description

P1: Quantum Desk Calculator

The task is to build quantum circuits that perform arithmetic operations between the binary representations of two positive integers. The circuit must be implemented in Qiskit and demonstrated on one or two instances.

P2: Quantum Phase Estimation

The primary function of the Quantum Phase Estimation (QPE) algorithm is to find an approximate value to an eigenvalue phase of some unitary matrix U , i.e. a real value $0 < \vartheta < 1$ such that

$$U |u\rangle = e^{2\pi i \vartheta} |u\rangle ,$$

for some eigenvector $|u\rangle$ of U .

The task is to implement the quantum circuit for QPE for the case where U is the controlled phase gate CR_φ with eigenvector $|11\rangle$.

Hint: you should know how the circuit is built for eigenvalues of the form $e^{2\pi i \vartheta}$; then take $\vartheta = \varphi/2\pi$.

P3: Quantum Counting

The Quantum Counting algorithm is a combination of the Grover algorithm and the QPE subroutine (cf. [P2](#)).

The task is to study the quantum circuit for this algorithm (cf. textbook), and apply it to the problem of finding a Hamiltonian cycle in a graph (HC problem). Recall that a Hamiltonian cycle of a graph is a simple cycle which visits every vertex of the graph. The HC problem is to determine whether a given graph has a Hamiltonian cycle or not. This problem belongs to the class of NP- complete problems, widely believed (but not yet proved) to be intractable on a classical computer.

By implementing in Qiskit a particular instance of this problem and the quantum counting circuit solving it, give evidence of the complexity of the quantum algorithm and compare it with the classical one.

P4: Breaking Diffie-Hellman

Explain the role of the Discrete Logarithm Algorithm (DLA) in an attack to the Diffie-Hellman key exchange protocol.

Define a quantum algorithm that exploits the Quantum Fourier Transform (QFT) to solve the DLA in polynomial time. Then implement it in Qiskit and demonstrate your implementation by showing how to break a simple instance of Diffie-Hellman.

P5: Your Proposal

Any project you would like to suggest (to be first discussed with me).